Safety Data Sheet

Sodium Hypochlorite 10%

1. Identification of the Substance or Mixture and of the Supplier

Identification:

- **Product name:** Sodium Hypochlorite 10%
- **Synonyms:** Antiformin, B-K liquid, Carrel-Dakin solution, Chloros, Chlorox, Clorox, Dakins solution, Deosan, Hyclorite, Hypochlorite solution containing > 7% available chlorine by wt. (UN 1791), Javex, Klorocin, Milton, Neo-cleaner, Neoseptal CL, Parozone, Purin B, Sodium chloride oxide, Sodium oxychloride, Surchlor
- **Formula:** NaOCl

Special identification: -

Relevant identified uses of the substance or mixture and uses advised against: Laboratory and industrial use

Details of the supplier of the safety data sheet:

- Prime Atosphere (Thailand) Co., Ltd.
- 888/66 Moo 3, Bang Pu Mai, Mueang Samut Prakan, Samut Prakan 10280, Thailand
- Tel: 064-654-4655
- Emergency Telephone Number: 064-654-4655

2. Hazards Identification

Classification of the substance/mixture (GHS):

- Skin corrosion/irritation: Category 1B
- Serious eye damage/eye irritation: Category 1
- Short-term (acute) aquatic hazard: Category 1
- Long-term (chronic) aquatic hazard: Category 1

Label Elements:



Signal word: Danger

Hazard statements:

- H372: Causes damage to the respiratory system through prolonged or repeated exposure
- H318: Causes serious eye damage
- H314: Causes severe skin burns and eye damage
- H400: Very toxic to aquatic life

Precautionary statements:

- P201: Obtain special instructions before use
- P261: Avoid breathing dust/fume/gas/mist/vapours/spray
- P264: Wash skin thoroughly after handling
- P280: Wear protective gloves/protective clothing/eye protection/face protection
- P301 + P330 + P331: IF SWALLOWED: Rinse mouth. Do NOT induce vomiting
- P304 + P340: IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing
- P303 + P361 + P353: IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water or shower
- P305 + P351 + P338: IF IN EYES: Rinse continuously with water for several minutes. Remove contact lenses if present and easy to do. Continue rinsing
- P403: Store in a well ventilated place
- P405: Store locked up
- P501: Dispose of contents/container to an approved waste disposal plant

Other hazards: -

3. Composition/Information on Ingredients

Chemical Name	CAS No.	Weight Percent (%)
Sodium Hypochlorite	7681-52-9	10
Water	7732-18-5	90

4. First Aid Measures

- **After swallowing:** Rinse mouth. Do NOT induce vomiting. Get medical attention immediately.
- **In case of skin contact:** Take off immediately all contaminated clothing. Rinse skin with plenty of water for at least 20 minutes. Get medical attention immediately.
- **After inhalation:** Remove to fresh air. If not breathing, give artificial respiration. Get medical attention immediately.
- **After eye contact:** Rinse with plenty of water. Remove contact lenses. Hold eyelids open and flush for at least 20 minutes. Get medical attention immediately.

5. Fire Fighting Measures

- **Unsuitable extinguishing media:** -
- **Suitable extinguishing media:** Water fog, Foam, Carbon dioxide (CO_2), Dry powder
- **Special hazards arising from the substance or mixture:** This substance is nonflammable. However, when heated, it decomposes to release toxic oxygen and chlorine gases.
- **Advice for firefighters:** Wear firefighting gear, chemical protective clothing and a self-contained breathing apparatus (SCBA). Use water spray or fog nozzles to cool containers and reduce vapour.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures: Evacuate the danger area (upwind). Remove ignition sources. Avoid inhalation of dusts/vapour. Avoid substance contact. Ensure adequate ventilation. Do not make sparks and flames.

Personal protection: Goggles/mask/gloves/chemical protective clothing

Environmental precautions: Do not let product enter drains.

Methods and materials for containment and cleaning up:

- Wear chemical protective clothing, a respirator with a chlorine gas filter cartridge, and goggles or a face shield.
- Ventilate the spill area.
- Use plastic tools to collect contaminated material. Prepare plastic bags and plastic drums (with lids).
- Put contaminated material in a plastic bag, tie it closed, place it in a plastic drum, close the lid and seal the lid edge with cloth tape.
- Label the drum “Chemical contaminated from accident” and dispose of it according to regulations.

7. Handling and Storage

Precautions for safe handling:

- Avoid prolonged exposure. Provide adequate ventilation in the work area. Wear suitable protective equipment. Do not contact the chemical directly.

Conditions for safe storage:

- Tightly closed
- Good ventilation
- Dry area
- Keep away from heat and sunlight

8. Exposure Controls/Personal Protection

Control parameters:

- STEL: 0.5 ppm
- TWA: 1 ppm

Suitable engineering control: Good ventilation system

Personal protective equipment:

- **Respiratory protection:** Filter mask
- **Eye/face protection:** Filter mask / breathing apparatus
- **Hand protection:** Gloves
- **Skin protection:** Chemical protective clothing
- **Hygiene measures:** Change clothing contaminated with chemicals. Wash hands and face after working with the substance and before eating, smoking or using the toilet. Do not eat, drink or smoke in the workplace.

9. Physical and Chemical Properties

- **Appearance:** Clear liquid
- **Color:** -
- **Odor:** Chlorine-like
- **Odor threshold:** -
- **pH:** 10.8 – 13.0
- **Melting point/freezing point:** -15 °C
- **Initial boiling point and boiling range:** > 40 °C
- **Flash point:** Non-flammable
- **Evaporation rate:** < 1
- **Flammability:** Non-flammable
- **Upper/lower flammability limits UEL/LEL (%):** -
- **Vapor pressure:** 12.1 mmHg at 20 °C
- **Vapor density (Air = 1):** -
- **Relative density (Water = 1):** 1.17 at 20 °C
- **Water solubility:** Soluble
- **Partition coefficient n-Octanol/Water (log Kow):** -
- **Autoignition temperature:** -
- **Decomposition temperature:** > 40 °C
- **Viscosity:** -

10. Stability and Reactivity

- **Reactivity:** Reacts violently or explosively with strong acids (e.g. Hydrochloric Acid, Nitric Acid), Acid compounds (e.g. Aluminium Chloride, Ferric Chloride, Alum), Acid based cleaning compounds (Brick, concrete cleaners), Ammonia Compounds (e.g. Ammonium Chloride, Ammonium Hydroxide, Quaternary Ammonium salts)
- **Chemical stability:** Stable under standard ambient conditions
- **Possibility of hazardous reactions:** No dangerous reaction known under conditions of normal use
- **Conditions to avoid:** Flame, spark, heat, moisture, sunlight
- **Incompatible materials:** Organic materials, Strong acids, Strong oxidizer, Ammonia, Metals, Amines, Ethyleneimine, Nitrogen, Reducing agents, Acids, Combustible material, Halogenated compounds, Metals, Metal salts, Zinc
- **Hazardous decomposition products:** See section 5

11. Toxicological Information

Acute Toxicity:

LC50	>5,000 mg/kg (>10% NaOCl) (Rat)
EC50	>10,500 mg/cm³ (Rat)
IC50	>10,000 mg/kg (Rabbit)

12. Ecological Information

Toxicity:

LC50	0.06 mg/l, Exposure time: 96 h (Rainbow Fish)
	5.9 mg/L, Exposure time: 96 h (Fathead minnow)
EC50	0.141 mg/l, Exposure time: 48 h (Water flea)
	0.035 mg/L, Exposure time: 48 h (Ceriodaphnia dubia)
IC50	0.023 mg/l, Exposure time: 7 d (Algae)

13. Disposal Considerations

- **Chemical and contaminated container disposal:** Disposal comply with local regulation

14. Transport Information

- **Road/Rail (ADR/RID):** UN 1791, HYPOCHLORITE SOLUTION, 8, III, Marine Pollutant
- **Air (IATA):** UN 1791, Hypochlorite solution, 8, III
- **Sea (IMDG):** UN 1791, HYPOCHLORITE SOLUTION, 8, III, Marine Pollutant

15. Regulatory Information

- Hazardous Substance Act B.E. 2535 (1992)
- Factory Act B.E. 2535 (1992)
- Notification of the Department of Industrial Works: Manual for Storage of Chemicals and Hazardous Substances B.E. 2550 (2007)
- Occupational Safety, Health and Environment Act B.E. 2554 (2011)

16. Other Information

- **Revised Date:** 10/07/2026